Appendix · When the compressors don't arrive

Team 4 · Liebherr CNc 5203 refrigerator, key component compressor · Sustainable Supply Chain Management, TH Köln, 2026

1 Key argument of each paper

Paper 1

Kleindorfer, P. R., & Saad, G. H. (2005). Managing disruption risks in supply chains. Production and Operations Management, 14(1), 53–68. https://doi.org/10.1111/j.1937-5956.2005.tb00009.x

The paper asks how firms can manage supply chain disruptions such as natural disasters, strikes or terrorism, which differ from normal supply and demand problems. It proposes the SAM framework (specify risk sources, assess, mitigate) and ten principles, among them that diversifying sourcing reduces risk and that extreme leanness makes a supply chain vulnerable, so some slack is needed; it supports this with US chemical industry accident data from 1995 to 2000. This explains our results: running lean on one far supplier without assessing the Cape risk caused 73 hours of line stoppage, while slack (8 extra days of safety stock) or a second sourcing option (Slovakia) removed almost all of it.

Paper 2

Bonilla, D., Keller, H., & Schmiele, J. (2015). Climate policy and solutions for green supply chains: Europe's predicament. Supply Chain Management: An International Journal, 20(3), 249–263. https://doi.org/10.1108/SCM-05-2014-0171

The authors ask whether moving production abroad really helps Europe meet its climate targets, and they answer with input–output tables for whole sectors, which follow the purchases between industries and so show the emissions behind a product wherever they occur. They find that if CO_2 is counted where it is emitted, offshoring looks like a cut in EU emissions, but if it is counted where the goods are used, it adds about 42 million t, roughly 12 % of the EU Kyoto target of 341 million t. Our Slovakia choice is the reverse move, and the 4.4 kg CO_2 saved per fridge fits that direction (about half from a cleaner grid when making the compressor, 16.5 vs 14.3 kg, and half from shorter transport, 2.7 vs 0.4 kg); because the paper works with sectors and not single components, we use it to support the direction of our result, not its size.

2 AI note

What we handed to AI. We used Claude (Anthropic) as a helper for most of the technical work. It wrote the first draft of the dataset and the texts in sources.csv, the notebook code, the code that fills results.json into the animation template, a first poster draft and short sheets that explained the code to us. We chose the product, the disruption and the countermeasure ourselves, looked up the road distances in Google Maps, found the factory videos and read both papers through the TH Köln library access.

How we checked it. Two of us went through the dataset row by row and asked for every sanity check where the comparison came from. We recalculated the derived numbers (distances, prices, lead times, CO2 per compressor) and ran the official check cell after every change.

AI mistakes we found. (1) The AI cited two scenes in the Liebherr factory video, at 6:09 and 8:55, that do not exist. We found this by watching the video at those times. (2) It wrote "route planners give the same order of magnitude" for the truck distances, but had never checked a route planner. We replaced the values with our own Google Maps distances. (3) It suggested Secop as the near supplier, but the sources showed that Secop no longer makes household compressors. We switched to Embraco Slovakia and marked it as an assumption.

What worked and what did not. AI was fast and reliable for code and formatting, and the check cell caught counting errors early. It was weak on facts: it filled gaps with claims that sounded right but were never checked, so every number needed a human check.

This note was drafted with Claude and checked and edited by our team.

3 Output of the official check cell

```
PASS  T1 Format
      all 10 files present, columns exact, no blanks or negatives
PASS  T2 Capacity
      WC1 Forming (thermoform inner liner, punch and bend sheet metal): busy 79.4% of available time
      WC2 Foaming (fill cabinet and doors with PU foam): busy 79.4% of available time
      WC3 Final assembly (compressor, cooling circuit, electronics, doors): busy 90.0% of available time
      WC4 Refrigerant charging and leak test: busy 75.0% of available time
      WC5 Quality test (test run, thermal camera): busy 71.1% of available time
      WC6 Packaging: busy 67.5% of available time
PASS  T3 Little's Law
      run 1: L = 12.40  vs  λ×W = 20.114/h × 0.62 h = 12.40  (gap 0.0%)
      10 run(s) checked
PASS  T4 Flow balance
      run 1: started 41,850, finished 41,837, key components used 41,837 (allowed 41,837 to 41,850)
PASS  T5 Margin
      cost per unit 292.74 to 293.28 EUR vs list price 713.45 EUR
--------------------------------------------------------------------------------
5 of 5 tests passed  ·  ALL PASS
check cell v1.0 · cell 3ad30383c119 · data af1963652edd · 2026-11-05 18:22
```

4 sources.csv (155 rows; long texts shortened here, full text in sources.csv of the dataset pack)

file	row_id	column	value	source_type	source_or_calculation	sanity_check
products.csv	LH-CNC5203	mass_kg	69.7	public	Coolblue product page, weight 69.7 kg: https://www.coolblue.de/…	Kueche24 lists 69.2 kg for the sister model CNd 5203-22, close…
products.csv	LH-CNC5203	list_price_eur	713.45	public	Recommended retail price 849 EUR incl. VAT on Coolblue (https:/…	Coolblue sells it for 756.90 EUR, so the list price is an upper…
bom.csv	CMP-COMPRESSOR	mass_kg	7.2	public	Embraco EMT30CDP (R600a piston compressor) net weight 7.2 kg: h…	household compressors typically weigh 6 to 10 kg
bom.csv	CMP-COMPRESSOR	unit_cost_eur	14.15	public	Huayi 2025: RMB 10.225 bn revenue / 87.83 m piston compressors…	average over all sizes; spare part prices in shops are much hig…
bom.csv	CMP-SHEET	unit_cost_eur	33.0	guesstim…	27.5 kg x 1.20 EUR/kg coated thin sheet incl. scrap	total material cost about 240 EUR = about one third of the net…
bom.csv	CMP-SHEET	mass_kg	27.5	guesstim…	rest of the appliance mass: 69.7 - 7.2 - 12 - 8 - 6 - 2 - 7 = 2…	sum of all parts without packaging = 69.7 kg = appliance mass
bom.csv	CMP-PLASTIC	unit_cost_eur	24.0	guesstim…	12 kg x 2.00 EUR/kg ABS/HIPS granulate	total material cost about 240 EUR = about one third of the net…
bom.csv	CMP-PLASTIC	mass_kg	12.0	guesstim…	estimate: inner liners for fridge and freezer part plus 3 drawe…	sum of all parts without packaging = 69.7 kg = appliance mass
bom.csv	CMP-FOAM	unit_cost_eur	22.4	guesstim…	8 kg x 2.80 EUR/kg PU system (polyol + isocyanate)	total material cost about 240 EUR = about one third of the net…
bom.csv	CMP-FOAM	mass_kg	8.0	guesstim…	estimate: about 0.2 m3 insulation foam x 40 kg/m3	sum of all parts without packaging = 69.7 kg = appliance mass
bom.csv	CMP-COOLKIT	unit_cost_eur	45.0	guesstim…	copper/aluminium tubes, condenser, evaporator, about 50 g R600a…	total material cost about 240 EUR = about one third of the net…
bom.csv	CMP-COOLKIT	mass_kg	6.0	guesstim…	estimate: condenser, evaporator, tubing	sum of all parts without packaging = 69.7 kg = appliance mass
bom.csv	CMP-ELECTRONICS	unit_cost_eur	55.0	guesstim…	control board with display, 1 to 2 fans for NoFrost, LED; estim…	total material cost about 240 EUR = about one third of the net…
bom.csv	CMP-ELECTRONICS	mass_kg	2.0	guesstim…	estimate: board, fans, wiring harness	sum of all parts without packaging = 69.7 kg = appliance mass
bom.csv	CMP-INTERIOR	unit_cost_eur	40.0	guesstim…	3 to 4 glass shelves, door bins, 2 door seals, hinges, handles;…	total material cost about 240 EUR = about one third of the net…
bom.csv	CMP-INTERIOR	mass_kg	7.0	guesstim…	estimate: glass shelves about 4 kg plus bins and fittings	sum of all parts without packaging = 69.7 kg = appliance mass
bom.csv	CMP-PACKAGING	unit_cost_eur	8.0	guesstim…	carton, EPS corners, film; estimate 8 EUR	total material cost about 240 EUR = about one third of the net…
bom.csv	CMP-PACKAGING	mass_kg	4.3	public	shipping weight 74 kg (Kueche24, sister model CNd 5203-22: http…	sum of all parts without packaging = 69.7 kg = appliance mass
bom.csv	CMP-COMPRESSOR	qty_per_unit	1.0	guesstim…	one set per refrigerator	one hermetic compressor per appliance; every fridge-freezer of…
bom.csv	CMP-SHEET	qty_per_unit	1.0	guesstim…	one set per refrigerator	one set = cabinet shell, side walls and two doors; its 27.5 kg…
bom.csv	CMP-PLASTIC	qty_per_unit	1.0	guesstim…	one set per refrigerator	one set = fridge and freezer inner liners plus 3 freezer drawer…
bom.csv	CMP-FOAM	qty_per_unit	1.0	guesstim…	one set per refrigerator	one foam fill per appliance (cabinet and both doors are foamed…
bom.csv	CMP-COOLKIT	qty_per_unit	1.0	guesstim…	one set per refrigerator	one condenser, one evaporator and one tube set per appliance, m…
bom.csv	CMP-ELECTRONICS	qty_per_unit	1.0	guesstim…	one set per refrigerator	one control board with display, NoFrost fan(s) and LED lighting…
bom.csv	CMP-INTERIOR	qty_per_unit	1.0	guesstim…	one set per refrigerator	one set of shelves, door bins, two door seals, hinges and handl…
bom.csv	CMP-PACKAGING	qty_per_unit	1.0	guesstim…	one set per refrigerator	one carton with EPS corners and film per appliance; 4.3 kg = sh…
workcenters.csv	WC1	num_servers	1.0	guesstim…	1 forming line with 1 operator	bottleneck is final assembly, as on the bike line
workcenters.csv	WC1	shift_hours	8.0	analogy	8 h as all bike stations	Liebherr runs several shifts; we model one shift of one line
workcenters.csv	WC1	days_per_week	5.0	analogy	5 days as bike data	Monday to Friday
workcenters.csv	WC1	availability	0.85	analogy	0.85 as bike WC1 (machine)	between 0 and 1
workcenters.csv	WC1	process_time_cv	0.15	analogy	bike WC1 has 0.2; forming is automated, so a bit less: 0.15	manual work varies more than machines
workcenters.csv	WC1	fixed_cost_eur_week	4480.0	guesstim…	labour cost 49.50 EUR/h in manufacturing 2025 (Destatis, https:…	labour part 1,980 EUR is public (Destatis); the machine part is…
routing.csv	WC1	process_time_min	2.0	guesstim…	guesstimate: 2.0 min per fridge for punching/bending the sheet…	WC1 capacity 25.5 per hour is only 13 % above the bottleneck: a…
routing.csv	WC1	setup_time_min	0.0	analogy	0 as bike data (setup is a week 3 bonus)	no batches in the model
process_environment.c…	WC1	kw_running	80.0	guesstim…	thermoforming heaters plus press: 80 kW running, 20 kW idle	gives about 6 kWh per refrigerator
process_environment.c…	WC1	kw_idle	20.0	guesstim…	thermoforming heaters plus press: 80 kW running, 20 kW idle	idle below running
process_environment.c…	WC1	water_l_per_unit	0.0	analogy	0 as bike data (column reserved)	-
workcenters.csv	WC2	num_servers	3.0	guesstim…	guesstimate: 3 foaming places because curing in the mould takes…	with 3 places WC2 stays below the bottleneck; with 2 places it…
workcenters.csv	WC2	shift_hours	8.0	analogy	8 h as all bike stations	Liebherr runs several shifts; we model one shift of one line
workcenters.csv	WC2	days_per_week	5.0	analogy	5 days as bike data	Monday to Friday
workcenters.csv	WC2	availability	0.85	analogy	0.85 as bike WC1 (machine)	between 0 and 1
workcenters.csv	WC2	process_time_cv	0.2	analogy	0.2 as bike WC1	manual work varies more than machines
workcenters.csv	WC2	fixed_cost_eur_week	2780.0	guesstim…	labour cost 49.50 EUR/h in manufacturing 2025 (Destatis, https:…	labour part 1,980 EUR is public (Destatis); the machine part is…
routing.csv	WC2	process_time_min	6.0	guesstim…	guesstimate: 6.0 min per fridge per foaming place, incl. curing…	WC2 capacity 25.5 per hour is only 13 % above the bottleneck: a…
routing.csv	WC2	setup_time_min	0.0	analogy	0 as bike data (setup is a week 3 bonus)	no batches in the model
process_environment.c…	WC2	kw_running	15.0	guesstim…	heated foaming mould plus dosing pump: 15 / 5 kW	gives about 6 kWh per refrigerator
process_environment.c…	WC2	kw_idle	5.0	guesstim…	heated foaming mould plus dosing pump: 15 / 5 kW	idle below running
process_environment.c…	WC2	water_l_per_unit	0.0	analogy	0 as bike data (column reserved)	-
workcenters.csv	WC3	num_servers	5.0	guesstim…	guesstimate: 5 assembly workplaces with one worker each, so tha…	bottleneck with 22.5 per hour; one workplace more (6) would mov…
workcenters.csv	WC3	shift_hours	8.0	analogy	8 h as all bike stations	Liebherr runs several shifts; we model one shift of one line
workcenters.csv	WC3	days_per_week	5.0	analogy	5 days as bike data	Monday to Friday
workcenters.csv	WC3	availability	0.9	analogy	0.90 as bike WC3 final assembly	between 0 and 1
workcenters.csv	WC3	process_time_cv	0.35	analogy	0.35 as bike WC3 final assembly	manual work varies more than machines
workcenters.csv	WC3	fixed_cost_eur_week	2180.0	guesstim…	labour cost 49.50 EUR/h in manufacturing 2025 (Destatis, https:…	labour part 1,980 EUR is public (Destatis); the machine part is…
routing.csv	WC3	process_time_min	12.0	guesstim…	guesstimate: 12.0 min of hands-on work per fridge per worker fo…	manual step with the most parts, so the longest time and the bo…
routing.csv	WC3	setup_time_min	0.0	analogy	0 as bike data (setup is a week 3 bonus)	no batches in the model
process_environment.c…	WC3	kw_running	1.5	guesstim…	cordless screwdrivers, conveyor, lights: 1.5 / 0.5 kW	gives about 6 kWh per refrigerator
process_environment.c…	WC3	kw_idle	0.5	guesstim…	cordless screwdrivers, conveyor, lights: 1.5 / 0.5 kW	idle below running
process_environment.c…	WC3	water_l_per_unit	0.0	analogy	0 as bike data (column reserved)	-
workcenters.csv	WC4	num_servers	2.0	guesstim…	2 charging places (R600a is flammable, charging and leak test t…	bottleneck is final assembly, as on the bike line
workcenters.csv	WC4	shift_hours	8.0	analogy	8 h as all bike stations	Liebherr runs several shifts; we model one shift of one line
workcenters.csv	WC4	days_per_week	5.0	analogy	5 days as bike data	Monday to Friday
workcenters.csv	WC4	availability	0.9	analogy	0.90 as bike WC3 and WC5	between 0 and 1
workcenters.csv	WC4	process_time_cv	0.25	analogy	between bike WC1 (0.2) and WC2 (0.3): 0.25	manual work varies more than machines
workcenters.csv	WC4	fixed_cost_eur_week	2680.0	guesstim…	labour cost 49.50 EUR/h in manufacturing 2025 (Destatis, https:…	labour part 1,980 EUR is public (Destatis); the machine part is…
routing.csv	WC4	process_time_min	4.0	guesstim…	guesstimate: 4.0 min per fridge to evacuate, charge R600a and l…	utilization at planned output 75 %, below 100 % (test T2)
routing.csv	WC4	setup_time_min	0.0	analogy	0 as bike data (setup is a week 3 bonus)	no batches in the model
process_environment.c…	WC4	kw_running	8.0	guesstim…	vacuum pump and leak detector: 8 / 3 kW	gives about 6 kWh per refrigerator
process_environment.c…	WC4	kw_idle	3.0	guesstim…	vacuum pump and leak detector: 8 / 3 kW	idle below running
process_environment.c…	WC4	water_l_per_unit	0.0	analogy	0 as bike data (column reserved)	-
workcenters.csv	WC5	num_servers	2.0	guesstim…	2 test places	bottleneck is final assembly, as on the bike line
workcenters.csv	WC5	shift_hours	8.0	analogy	8 h as all bike stations	Liebherr runs several shifts; we model one shift of one line
workcenters.csv	WC5	days_per_week	5.0	analogy	5 days as bike data	Monday to Friday
workcenters.csv	WC5	availability	0.95	analogy	0.95 as bike WC4 quality inspection	between 0 and 1
workcenters.csv	WC5	process_time_cv	0.5	analogy	0.5 as bike WC4 quality inspection	manual work varies more than machines
workcenters.csv	WC5	fixed_cost_eur_week	2280.0	guesstim…	labour cost 49.50 EUR/h in manufacturing 2025 (Destatis, https:…	labour part 1,980 EUR is public (Destatis); the machine part is…
routing.csv	WC5	process_time_min	4.0	guesstim…	guesstimate: 4.0 min per fridge for thermal image, visual check…	utilization at planned output 71 %, below 100 % (test T2); note…
routing.csv	WC5	setup_time_min	0.0	analogy	0 as bike data (setup is a week 3 bonus)	no batches in the model
process_environment.c…	WC5	kw_running	3.0	guesstim…	appliances running in the test plus camera: 3 / 0.5 kW	gives about 6 kWh per refrigerator
process_environment.c…	WC5	kw_idle	0.5	guesstim…	appliances running in the test plus camera: 3 / 0.5 kW	idle below running
process_environment.c…	WC5	water_l_per_unit	0.0	analogy	0 as bike data (column reserved)	-
workcenters.csv	WC6	num_servers	1.0	guesstim…	1 packaging place	bottleneck is final assembly, as on the bike line
workcenters.csv	WC6	shift_hours	8.0	analogy	8 h as all bike stations	Liebherr runs several shifts; we model one shift of one line
workcenters.csv	WC6	days_per_week	5.0	analogy	5 days as bike data	Monday to Friday
workcenters.csv	WC6	availability	0.9	analogy	0.90 as bike WC5 packaging	between 0 and 1
workcenters.csv	WC6	process_time_cv	0.2	analogy	0.2 as bike WC5 packaging	manual work varies more than machines
workcenters.csv	WC6	fixed_cost_eur_week	2180.0	guesstim…	labour cost 49.50 EUR/h in manufacturing 2025 (Destatis, https:…	labour part 1,980 EUR is public (Destatis); the machine part is…
routing.csv	WC6	process_time_min	1.8	guesstim…	guesstimate: 1.8 min per fridge. Liebherr video 'Wie entsteht e…	automated step, so short and the least loaded station (68 %)
routing.csv	WC6	setup_time_min	0.0	analogy	0 as bike data (setup is a week 3 bonus)	no batches in the model
process_environment.c…	WC6	kw_running	5.0	guesstim…	wrapping and strapping machine: 5 / 1 kW	gives about 6 kWh per refrigerator
process_environment.c…	WC6	kw_idle	1.0	guesstim…	wrapping and strapping machine: 5 / 1 kW	idle below running
process_environment.c…	WC6	water_l_per_unit	0.0	analogy	0 as bike data (column reserved)	-
process_environment.c…	WC1	waste_kg_per_unit	3.0	public	Liebherr plant Marica 2024: 4.07 kg waste per appliance (https:…	sum 4.07 kg per appliance
process_environment.c…	WC2	waste_kg_per_unit	0.6	public	Liebherr plant Marica 2024: 4.07 kg waste per appliance (https:…	sum 4.07 kg per appliance
process_environment.c…	WC3	waste_kg_per_unit	0.2	public	Liebherr plant Marica 2024: 4.07 kg waste per appliance (https:…	sum 4.07 kg per appliance
process_environment.c…	WC4	waste_kg_per_unit	0.0	public	Liebherr plant Marica 2024: 4.07 kg waste per appliance (https:…	sum 4.07 kg per appliance
process_environment.c…	WC5	waste_kg_per_unit	0.0	public	Liebherr plant Marica 2024: 4.07 kg waste per appliance (https:…	sum 4.07 kg per appliance
process_environment.c…	WC6	waste_kg_per_unit	0.27	public	Liebherr plant Marica 2024: 4.07 kg waste per appliance (https:…	sum 4.07 kg per appliance
locations.csv	LOC-PLANT	lat	48.07222	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	town centre of Ochsenhausen; the Liebherr plant (Memminger Stra…
locations.csv	LOC-PLANT	lon	9.94806	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	town centre of Ochsenhausen; the Liebherr plant (Memminger Stra…
locations.csv	LOC-SUP-CN	lat	30.747	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	seat of the Jiaxing city government; Jiaxipera is in Nanhu dist…
locations.csv	LOC-SUP-CN	lon	120.756	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	seat of the Jiaxing city government; Jiaxipera is in Nanhu dist…
locations.csv	LOC-SUP-SK	lat	48.94389	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	town centre of Spisska Nova Ves; plant location inside the town…
locations.csv	LOC-SUP-SK	lon	20.5675	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	town centre of Spisska Nova Ves; plant location inside the town…
locations.csv	LOC-SHA	lat	30.626539	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	port coordinates (Yangshan deep-water port, 30.6 N 122.1 E), no…
locations.csv	LOC-SHA	lon	122.0649…	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	port coordinates (Yangshan deep-water port, 30.6 N 122.1 E), no…
locations.csv	LOC-RTM	lat	51.885	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	port coordinates of Rotterdam (51.9 N 4.3 E), not a town centre
locations.csv	LOC-RTM	lon	4.2867	public	coordinates of the town or port on Wikipedia: https://en.wikipe…	port coordinates of Rotterdam (51.9 N 4.3 E), not a town centre
network_lanes.csv	LANE-CN-SHA	distance_km	91.0	public	Google Maps, fastest car route Jiaxing to Yangshan deep-water p…	our first estimate (straight line x 1.25) was 157 km; the motor…
network_lanes.csv	LANE-CN-SHA	transit_days_mean	0.5	guesstim…	91 km to the port: half a day	Google Maps: 1 h 15 min by car; half a day includes loading and…
network_lanes.csv	LANE-CN-SHA	transit_days_sd	0.2	analogy	bike LANE-RTM-PLANT has 0.2 days; kept at 0.2	40 % of the 0.5-day mean, because queues at the port gate vary…
network_lanes.csv	LANE-SHA-RTM	distance_km	19631.0	public	10,600 nm via Suez (https://porteconomicsmanagement.org/routing…	bike Taiwan-Rotterdam has 20,000 km
network_lanes.csv	LANE-SHA-RTM	transit_days_mean	35.0	public	Asia-Europe service via Suez with port calls about 35 days (htt…	pure sailing 27 days
network_lanes.csv	LANE-SHA-RTM	transit_days_sd	3.0	analogy	3 days as bike LANE-EAST-RTM	about 10 % of the mean
network_lanes.csv	LANE-SHA-RTM-CAPE	distance_km	25558.0	public	13,800 nm around the Cape (https://porteconomicsmanagement.org/…	about 30 % longer than Suez
network_lanes.csv	LANE-SHA-RTM-CAPE	transit_days_mean	43.0	public	35 days + 8 days detour (27 vs. 35 days pure sailing, https://p…	route used in the disruption scenario (Red Sea 2024)
network_lanes.csv	LANE-SHA-RTM-CAPE	transit_days_sd	5.0	guesstim…	more spread, because schedules were disrupted	larger than the Suez lane
network_lanes.csv	LANE-RTM-PLANT	distance_km	778.0	public	Google Maps, fastest car route Port of Rotterdam (Europoort) to…	our first estimate was 732 km (6 % too low); the route via A3 i…
network_lanes.csv	LANE-RTM-PLANT	transit_days_mean	1.5	guesstim…	778 km: a good day of driving plus pick-up	Google Maps: 8.5 h by car; a truck needs about 10 h, more than…
network_lanes.csv	LANE-RTM-PLANT	transit_days_sd	0.3	analogy	bike LANE-RTM-PLANT: 0.2 on 1 day; a bit more here	20 % of the mean, the same share as bike LANE-RTM-PLANT (0.2 on…
network_lanes.csv	LANE-SK-PLANT	distance_km	983.0	public	Google Maps, fastest car route Spisska Nova Ves to Memminger St…	our first estimate was 985 km, almost the same; the route via t…
network_lanes.csv	LANE-SK-PLANT	transit_days_mean	1.5	guesstim…	983 km: 1.5 days with driving-time rules	Google Maps: 10.5 h by car; a truck needs about 13 h = 9 h on d…
network_lanes.csv	LANE-SK-PLANT	transit_days_sd	0.3	analogy	bike LANE-RTM-PLANT: 0.2 on 1 day; a bit more here	20 % of the mean, the same share as bike LANE-RTM-PLANT (0.2 on…
freight_rates.csv	LANE-CN-SHA	rate_eur_per_shipm…	120.0	analogy	as bike LANE-RTM-PLANT (truck from the port)	Chinese truck rates are probably lower than European; small eff…
freight_rates.csv	LANE-CN-SHA	rate_eur_per_tkm	0.1	analogy	as bike LANE-RTM-PLANT (truck from the port)	5.8 t x 91 km x 0.10 = about 53 EUR per order on top of the fix…
freight_rates.csv	LANE-CN-SHA	min_charge_eur	90.0	analogy	as bike LANE-RTM-PLANT (truck from the port)	below the fixed rate of 120 EUR, so it never applies
freight_rates.csv	LANE-RTM-PLANT	rate_eur_per_shipm…	120.0	analogy	as bike LANE-RTM-PLANT (truck from the port)	same leg type as in the bike data (port to plant by truck)
freight_rates.csv	LANE-RTM-PLANT	rate_eur_per_tkm	0.1	analogy	as bike LANE-RTM-PLANT (truck from the port)	5.8 t x 778 km x 0.10 = about 451 EUR per order; a full truck R…
freight_rates.csv	LANE-RTM-PLANT	min_charge_eur	90.0	analogy	as bike LANE-RTM-PLANT (truck from the port)	below the fixed rate of 120 EUR, so it never applies
freight_rates.csv	LANE-SK-PLANT	rate_eur_per_shipm…	150.0	analogy	as bike LANE-NORTH-PLANT (truck straight from the supplier)	supplier straight to plant, like bike Frames North
freight_rates.csv	LANE-SK-PLANT	rate_eur_per_tkm	0.09	analogy	as bike LANE-NORTH-PLANT (truck straight from the supplier)	weekly order 5.8 t x 983 km x 0.09 = about 513 EUR per order on…
freight_rates.csv	LANE-SK-PLANT	min_charge_eur	100.0	analogy	as bike LANE-NORTH-PLANT (truck straight from the supplier)	below the fixed rate of 150 EUR, so it never applies
freight_rates.csv	LANE-SHA-RTM	rate_eur_per_shipm…	3091.06	public	Drewry WCI Shanghai-Rotterdam 22 Oct 2026: 3,632 USD per 40 ft…	weekly order about 810 compressors x 7.2 kg = 5.8 t, below the…
freight_rates.csv	LANE-SHA-RTM	rate_eur_per_tkm	0.0	guesstim…	container rate is a flat price, not by weight	0 is allowed
freight_rates.csv	LANE-SHA-RTM	min_charge_eur	3091.06	guesstim…	at least one full container	= container rate
freight_rates.csv	LANE-SHA-RTM-CAPE	rate_eur_per_shipm…	5292.77	public	Drewry WCI Shanghai-Rotterdam 5 Sep 2024 (Red Sea crisis): 6,21…	Drewry: +158 % on the year before (DHL article)
freight_rates.csv	LANE-SHA-RTM-CAPE	rate_eur_per_tkm	0.0	guesstim…	flat price per container	0 is allowed
freight_rates.csv	LANE-SHA-RTM-CAPE	min_charge_eur	5292.77	guesstim…	at least one full container	= container rate
suppliers.csv	SUP-CN	unit_price_eur	14.15	public	same as bom.csv CMP-COMPRESSOR (Huayi revenue / units 2025)	cheaper than Europe
suppliers.csv	SUP-SK	unit_price_eur	19.64	analogy	China price x 1.39; 1.39 = bike Frames Local 118 EUR / Frames E…	near is dearer than far, as in the bike data
suppliers.csv	SUP-CN	promised_lead_days	32.0	analogy	bike: promise 30 with actual 35; here 36 / 1.17 = about 32	smaller than the actual mean
suppliers.csv	SUP-CN	lead_time_mean_days	36.0	guesstim…	calendar days: production 7 + truck 0.5 + port Shanghai 3 + sea…	bike Taiwan: 35 working days
suppliers.csv	SUP-CN	lead_time_sd_days	3.5	guesstim…	sea spread 3 calendar days plus port congestion, about 10 % of…	bike Taiwan: 6 on 35
suppliers.csv	SUP-SK	promised_lead_days	5.0	analogy	5 as bike Frames North	truck 1.5 days plus production
suppliers.csv	SUP-SK	lead_time_mean_days	5.5	analogy	bike Frames North 6.5; the Slovakia lane is short: 5.5	larger than the promise
suppliers.csv	SUP-SK	lead_time_sd_days	1.5	guesstim…	half of bike Frames North (3), because truck only	smaller than the mean
suppliers.csv	SUP-CN	capacity_per_month	5000.0	guesstim…	our share: Jiaxipera makes 30 m compressors per year (https://z…	covers our need of about 3,500 per month
suppliers.csv	SUP-SK	capacity_per_month	4500.0	guesstim…	guesstimate of spare capacity. Embraco moved its Turin producti…	ASSUMPTION: that this plant still makes household compressors i…
suppliers.csv	SUP-CN	min_per_month	0.0	guesstim…	no minimum order agreed	0 means the model may order any amount
suppliers.csv	SUP-SK	min_per_month	0.0	guesstim…	no minimum order agreed	0 means the model may order any amount; a real switch would lik…
suppliers.csv	SUP-CN	kg_co2e_per_unit	16.5	guesstim…	7.2 kg x 1.92 kg CO2/kg steel (worldsteel 2024, https://worldst…	production only, transport comes from the lanes; simplification…
suppliers.csv	SUP-SK	kg_co2e_per_unit	14.3	guesstim…	7.2 kg x 1.92 kg CO2/kg steel (worldsteel 2024) + 5 kWh product…	same simplification as SUP-CN; lower only because of the Slovak…
emission_factors.csv	crude_steel_global_…	value	1.92	public	worldsteel Sustainability Indicators 2025: https://worldsteel.o…	world average, unchanged since 2021
emission_factors.csv	grid_electricity_CN…	value	0.525	public	OWID: 525 g CO2/kWh China 2025 (https://ourworldindata.org/grap…	coal power, high value
emission_factors.csv	grid_electricity_SK…	value	0.095	public	OWID: 95 g CO2/kWh Slovakia 2025 (https://ourworldindata.org/gr…	much nuclear power, low value